Logarithms and Their Properties

Section 1: Definition of Logarithm and Inverses

1 Convert $\log_2(32) = 5$ to its equivalent exponential form.

a. $2^5 = 32$

b. $5^2 = 32$

c. $32^{1/5} = 2$

d. $2 \times 5 = 10$

2 Evaluate $\log_3\left(\frac{1}{27}\right)$.

a. -3

b. 3

c. $1/3$

d. $-1/3$

3 What is the inverse function of $f(x) = e^{2x-1}$?

a. $f^{-1}(x) = \frac{\ln x + 1}{2}$

b. $f^{-1}(x) = \ln(2x - 1)$

c. $f^{-1}(x) = 2\ln x - 1$

d. $f^{-1}(x) = e^{-(2x-1)}$

Section 2: Product, Quotient, and Power Rules

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4 Expand $\log_b\left(\frac{x^3y}{z^2}\right)$ into individual logarithms.

a. $3\log_b x + \log_b y - 2\log_b z$

b. $3\log_b x \cdot \log_b y - 2\log_b z$

c. $6(\log_b x + \log_b y - \log_b z)$

d. $\frac{3\log_b x + \log_b y}{2\log_b z}$

5 Condense into a single logarithm: $2\ln(x) + \frac{1}{2}\ln(y) - 3\ln(z)$.

a. $\ln\left(\frac{x^2\sqrt{y}}{z^3}\right)$

b. $\ln(x^2 + \sqrt{y} - z^3)$

c. $\ln\left(\frac{2x \cdot \frac{1}{2}y}{3z}\right)$

d. $\ln(x^2y^{1/2}z^3)$

6 Given $\log_a 2 \approx 0.301$ and $\log_a 3 \approx 0.477$, approximate $\log_a 18$.

a. 1.255

b. 0.778

c. 1.079

d. 0.954

Section 3: Change of Base Formula

7 State the change of base formula to express $\log_b a$ in terms of natural logarithms.

a. $\log_b a = \frac{\ln a}{\ln b}$

b. $\log_b a = \frac{\ln b}{\ln a}$

c. $\log_b a = \ln(a - b)$

d. $\log_b a = \ln(a) \cdot \ln(b)$

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8 Evaluate $\log_4(32)$ using the change of base formula with base 2.

a. $5/2 = 2.5$

b. 8

c. 2

d. $5/4$

9 Evaluate the product $\log_2 3 \cdot \log_3 4 \cdot \log_4 8$.

a. 3

b. 2

c. 4

d. $\log_2 8$ is 3

Section 4: Domains and Graphs of Logarithmic Functions

10 Find the domain and vertical asymptote of $f(x) = \ln(2x - 6)$.

a. Domain: $(3, \infty)$, Asymptote: $x = 3$

b. Domain: $[3, \infty)$, Asymptote: $x = 3$

c. Domain: $(6, \infty)$, Asymptote: $x = 6$

d. Domain: $(0, \infty)$, Asymptote: $x = 0$

11 What is the range of any basic logarithmic function $f(x) = \log_b(x)$ $(b > 1)$?

a. $(-\infty, \infty)$

b. $(0, \infty)$

c. $[0, \infty)$

d. $(-\infty, 0)$

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12 Find the x-intercept of $y = \log_3(x + 2) - 1$.

a. $(1, 0)$

b. $(0, 0)$

c. $(3, 0)$

d. $(-1, 0)$

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